Your notes

Properties of Waves

- **Travelling waves** are defined as:

Oscillations that transfer energy from one place to another without transferring matter

- Waves transfer **energy**, **not** matter
- Waves are generated by **oscillating sources**
  - These oscillations travel **away** from the source
- Oscillations can propagate through a **medium** (e.g. air, water) or in a **vacuum** (i.e. no particles), depending on the type of wave
- The key properties of **travelling waves** are:
  - displacement
  - wavelength
  - amplitude
  - period
  - frequency
  - wave speed

Displacement

- Displacement x of a wave is the distance of a point on the wave from its equilibrium position
  - It is a vector quantity; it can be positive or negative
  - Measured in **metres** (m)

Wavelength

- Wavelength λ is the length of one complete oscillation measured from the same point on two consecutive waves
  - For example, two crests, or two troughs
  - Measured in **metres** (m)

Amplitude

- Amplitude A is the maximum displacement of an oscillating wave from its equilibrium position ($x = 0$)
  - Amplitude can be positive or negative depending on the direction of the displacement

- Measured in **metres** (m)
- Where the wave has 0 amplitude (the horizontal line) is referred to as the **equilibrium position**

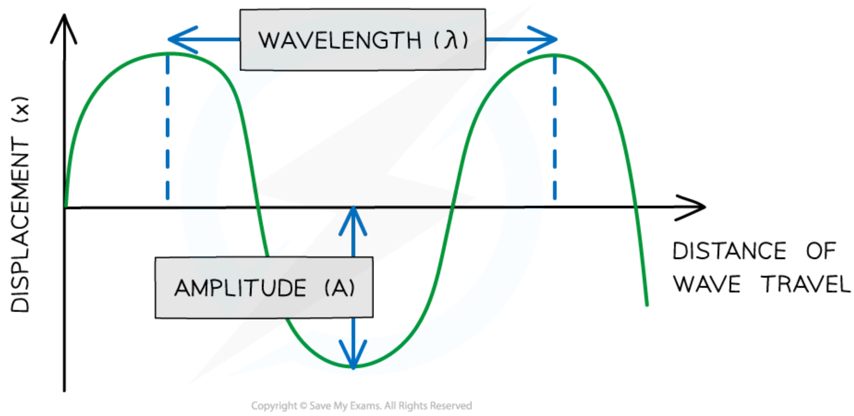


Diagram showing the amplitude and wavelength of a wave

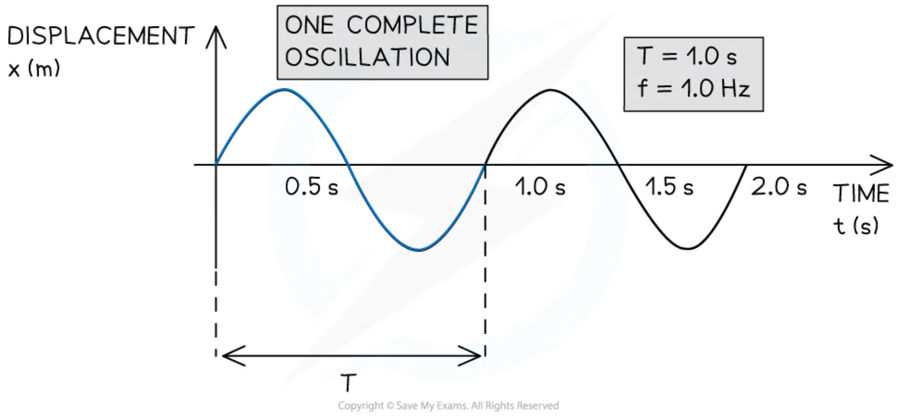
Period and frequency

- **Period** (*T*) is the time taken for a complete oscillation to pass a fixed point
  - Measured in **seconds** (s)
- **Frequency** (*f*) is the number of complete oscillations to pass a fixed point per second
  - Measured in **Hertz** (Hz)
- The frequency, *f*, and the period, *T*, of a travelling wave are related to each other by the equation:

$$f = \frac{1}{T}$$

- Where:
  - f = frequency (Hz)
  - T = time period (s)

Your notes



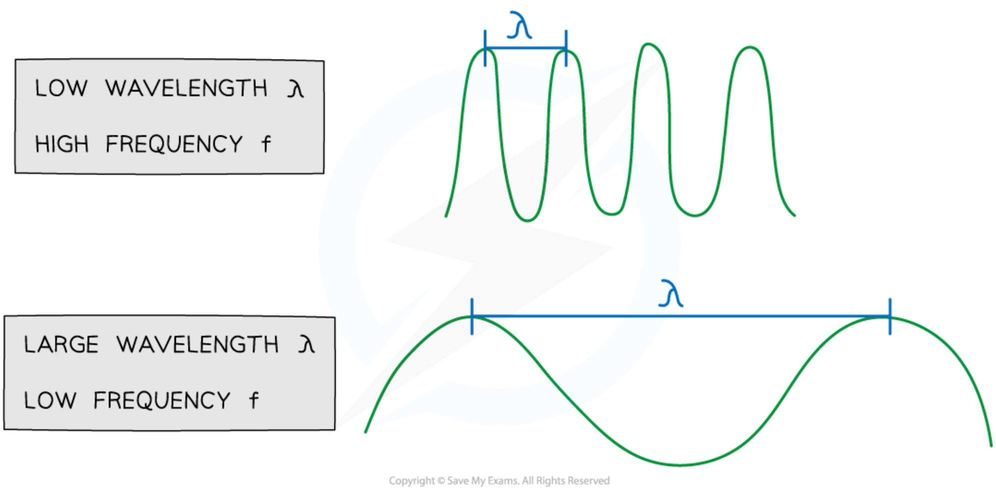
Period T and frequency f of a travelling wave

Wave speed

- **Wave speed** (v) is the distance travelled by the wave per unit time
  - Measured in **metres per second** ($m\ s^{-1}$)
- The wave speed is defined by the equation:

$$v = f\lambda = \frac{\lambda}{T}$$

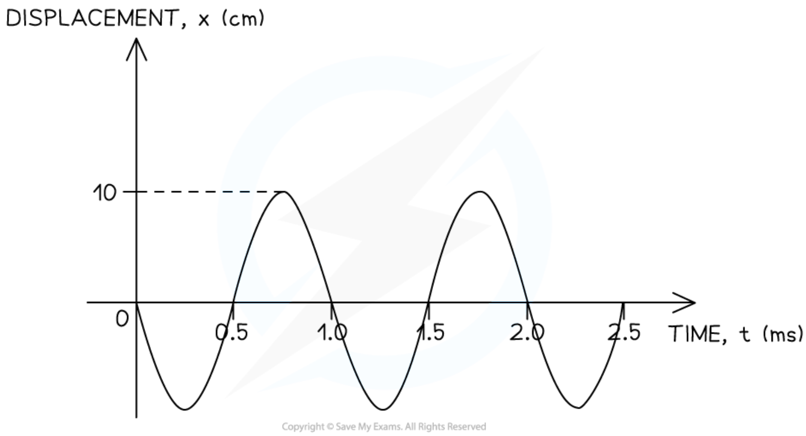
- Where:
  - v = wave speed ($m\ s^{-1}$)
  - λ = wavelength (m)
- This is referred to as the **wave equation**
- It tells us that for a wave of constant speed:
  - As the wavelength **increases,** the frequency **decreases**
  - As the wavelength **decreases,** the frequency **increases**



The relationship between the frequency and wavelength of a wave

Worked Example

The graph below shows a travelling wave.



Determine:

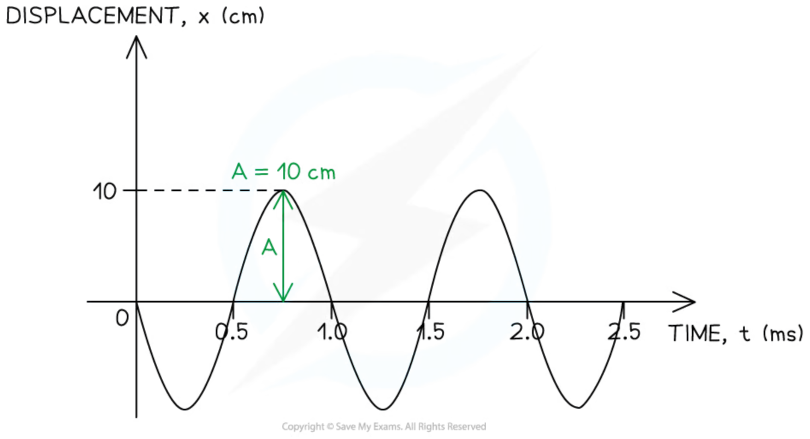
(a) The amplitude *A* of the wave, in m.

(b) The frequency *f* of the wave, in Hz.

Answer:

(a) Identify the amplitude *A* of the wave on the graph

- The amplitude is defined as the maximum displacement from the equilibrium position ($x = 0$)



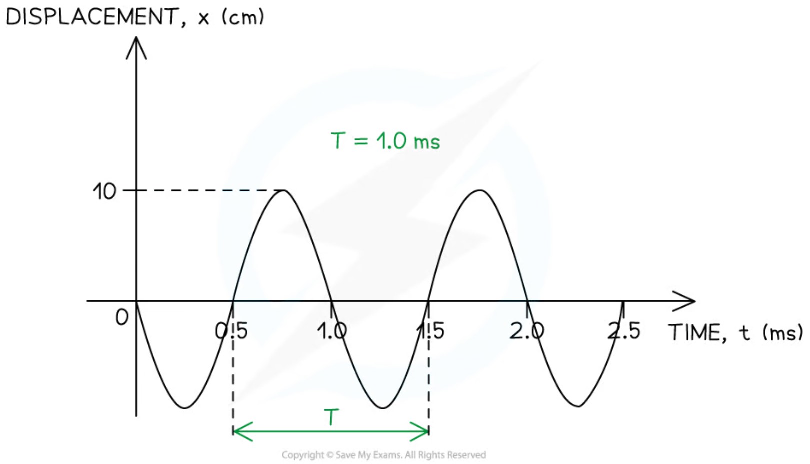
- The amplitude must be converted from centimetres (cm) into metres (m)

$A = 0.1$ m

(b) Calculate the frequency of the wave

Step 1: Identify the period *T* of the wave on the graph

- The period is defined as the time taken for one complete oscillation to occur



- The period must be converted from milliseconds (ms) into seconds (s)

$$T = 1 \times 10^{-3} \text{ s}$$

Step 2: Write down the relationship between the frequency *f* and the period *T*

$$f = \frac{1}{T}$$

Step 3: Substitute the value of the period determined in Step 1

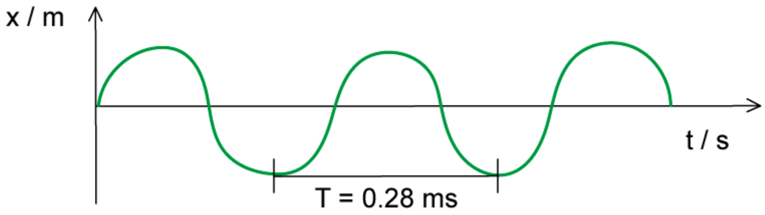
$$f = \frac{1}{1 \times 10^{-3}} = 1000 \text{ Hz}$$

Your notes

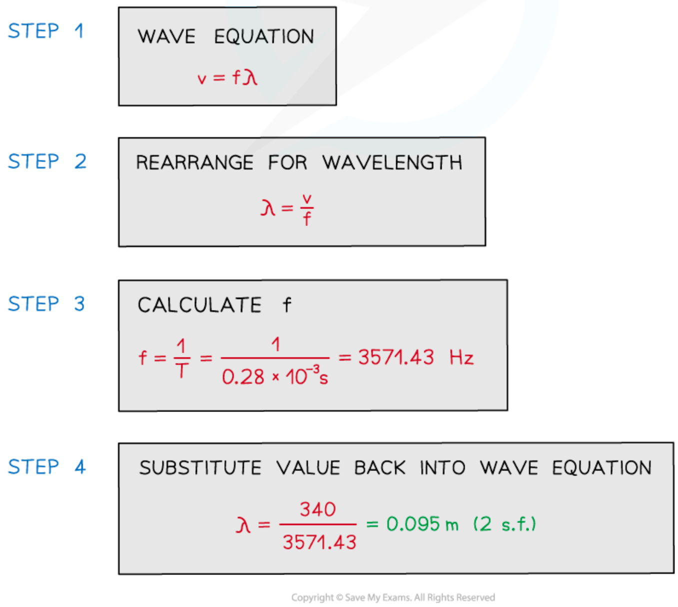
Worked Example

The wave in the diagram below has a speed of 340 m s^{-1}.

Determine the wavelength of the wave.



Answer:



Worked Example

A travelling wave has a period of 1.0 μs and travels at a velocity of 100 cm s^{-1}.

Calculate the wavelength of the wave, in m.

Answer:

Step 1: Write down the known quantities

- Period, $T = 1.0\ \mu s = 1.0 \times 10^{-6}$ s
- Velocity, $v = 100$ cm $s^{-1} = 1.0$ m s^{-1}

Note the conversions:

- The period must be converted from microseconds (μs) into seconds (s)

- The velocity must be converted from cm s^{-1} into m s^{-1}

Step 2: Write down the relationship between the frequency *f* and the period *T*

$$f = \frac{1}{T}$$

Step 3: Substitute the value of the period into the above equation to calculate the frequency

$$f = \frac{1}{1 \times 10^{-6}} = 1 \times 10^{6}$$

Step 4: Write down the wave equation

$$v = f\lambda$$

Step 5: Rearrange the wave equation to calculate the wavelength λ

$$\lambda = \frac{v}{f}$$

Step 6: Substitute the numbers into the above equation

$$\lambda = \frac{1.0}{1.0 \times 10^{6}} = 1.0 \times 10^{-6} \text{ m}$$

Examiner Tips and Tricks

You must be able to interpret different properties of waves from a variety of graphs. You may recognise that calculating the time period and wavelength looks very similar (the distance for one full wave). This is the time period if the *x-axis* is **time**. If the *x*-axis is **distance**, then this distance is the wavelength.

Pay very close attention to units. If you want a frequency in Hertz, then the time period must be in **seconds** and not milliseconds.

Your notes